

Clever Hans: Testing Competing Explanations

A case-based lesson for teaching skepticism, hypothesis formation, experimental design, evidence, and inference.

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LEARNING OUTCOMES

Formulate Turn a broad claim into testable hypotheses and alternatives.	Design Identify manipulations that distinguish among competing explanations.	Infer Use results to decide what the evidence supports - and what it does not.
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1

START WITH THE CLAIM

Can a horse answer questions?

Students are introduced to the historical claim that Clever Hans could answer questions by tapping his hoof. Reported abilities included arithmetic, telling time, identifying musical intervals, and naming people.

Ask before explaining: What exactly is being claimed? What would you need to observe before accepting that claim?

2

MAKE THE CLAIM TESTABLE

What are the competing explanations?

The question "Is Hans clever?" is too vague to test well. Students translate it into explanations that make different predictions.

Explanation A Hans independently solves the question and communicates the answer by tapping.	Explanation B Hans has learned a routine or responds to features of the testing situation.	Explanation C Hans uses information unintentionally provided by the human questioner.
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Student task: For each explanation, write one result that would make it more plausible and one result that would make it less plausible.

3

DESIGN DISCRIMINATING TESTS

Change one source of information at a time.

The investigation described in the case varied several features of the interaction. Students predict the outcome expected under each explanation before seeing the results.

Manipulation	Why change it?	Prediction to write before results
Different questioners	Tests whether performance depends specifically on Hans's owner.	Would independent problem solving require one particular person?
Questioner knows vs. does not know the answer	Removes information the questioner could unintentionally transmit.	What should happen if Hans calculates independently?
Vary distance	Changes access to subtle visual information.	Should distance matter to arithmetic ability?
Restrict visual contact	Directly limits visual information from the questioner.	What should happen under each explanation?

REVEAL THE EVIDENCE

4

What pattern do the results produce?

Visual access mattered.

Hans needed visual contact with the questioner; greater distance reduced accuracy, and blinkering reduced it further.

Questioner knowledge mattered.

Hans answered correctly when the questioner knew the answer, but not when the questioner did not know it.

Ask: Which explanation predicts both results? Which explanations are now difficult to maintain?

BUILD THE INFERENCE

5

The best explanation is not the most dramatic one.

The evidence supports the conclusion that Hans was responding to visual cues from the questioner rather than independently working out the answers. The cues did not need to be consciously produced; subtle changes in posture, expression, or tension could provide information about when to stop tapping.

Inference check: Does this evidence show that Hans was unintelligent? No. It addresses a narrower question: what information explained his correct responses in this task?

TRANSFER THE REASONING

6

Why does blinding matter?

The case provides a concrete route into experimenter effects and blinding. Students identify other situations in which expectations could unintentionally influence a subject, measurement, or interpretation.

Core lesson

Scientific skepticism is productive when it generates better tests. The goal is not to reject surprising claims automatically, but to design conditions under which competing explanations make different predictions.

STUDENT WORKSHEET / DISCUSSION PROMPTS

1. Rewrite the claim "Hans is clever" as a testable hypothesis.
2. List at least two alternative explanations for Hans's performance.
3. Choose one manipulation from the case. What outcome would you predict if Hans solved the questions independently?
4. Which result most strongly changes your interpretation, and why?
5. What conclusion is supported by the evidence? Write it without claiming more than the experiment shows.
6. Design one additional test that would further separate the remaining explanations.

INSTRUCTOR SYNTHESIS

Skepticism

What else could explain the observation?

Empiricism

What evidence would distinguish among those explanations?

Experimental control

What source of information can we remove or manipulate?

Scientific inference

What does the evidence justify concluding - and what remains unresolved?